

PROJECT REPORT

ATM SIMULATOR

A simple console-based banking application using Python

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1. Introduction

For this project, I developed a basic ATM Simulator using Python. The program runs in the console and allows a user to perform a few common banking operations, such as checking a balance, depositing money, withdrawing money, and viewing a mini statement. It also includes a PIN check and an option to change the PIN.

I chose this project because it gave me a chance to practise the Python topics covered in class and understand how separate functions and files can work together in one program. This is a learning project, so it does not connect to a real bank or store information permanently.

2. Project Objectives

The main objectives of this project are:

- To create a simple ATM menu that is easy to use.
- To check a user's PIN before allowing access to the menu.
- To support balance checking, deposits, and withdrawals.
- To keep a list of transactions during the current program session.
- To handle common incorrect inputs without stopping the program unexpectedly.
- To organize the program into small Python modules that are easier to understand and maintain.

3. Project Scope

The simulator is designed for basic practice. It starts with a sample balance of Rs. 5,000 and a default PIN of 1234. The user can try the available operations until choosing the Exit option. The balance, PIN, and transaction history are stored in the program while it is running. When the program is closed, these values return to their initial state the next time it is started.

The program is not intended for real financial use. It does not connect to a bank, use a database, or provide real account security.

4. Tools and Python Concepts Used

Language: Python 3

Interface: Command-line / console

External libraries: None required

The project uses variables to store the balance and PIN, input and print statements to communicate with the user, conditional statements to select operations, and loops to keep the menu running. Functions divide the work into smaller tasks. A list stores the transaction history, while dictionaries hold the details of each transaction. The program also uses modules and imports to keep related functions in separate files. A try-except block handles invalid numeric input.

5. Project Structure

File	Purpose
main.py	Starts the program, manages login and the menu, and connects the other modules.
authentication.py	Checks the entered PIN and limits the user to three attempts.
menu.py	Displays the list of ATM operations.
account.py	Handles balance checking, deposits, and withdrawals.
transactions.py	Adds transactions to the history list and displays the mini statement.
pin_management.py	Checks the current PIN and allows the user to set a new four-digit PIN.

6. How the Program Works

1. The program displays a welcome message.
2. The user enters the four-digit PIN. If it is incorrect, the program allows another attempt, up to three attempts in total.
3. After a successful login, the ATM menu is shown.
4. The user chooses an option by entering its number.
5. The program performs the selected task and shows the result.
6. After completing an operation, the menu is displayed again.
7. The program ends when the user selects Exit, or when all three login attempts are incorrect.

7. Main Features

PIN authentication: The user must enter the correct PIN to access the menu. Three incorrect attempts block access for that run.

Check balance: Displays the current available balance.

Deposit money: Adds a positive amount to the balance. If the input is not a valid number or is not greater than zero, the program displays an error and leaves the balance unchanged.

Withdraw money: Allows withdrawals in multiples of Rs. 100, provided the amount is positive and does not exceed the available balance.

Mini statement: Shows the transactions made during the current session, including their type, amount, and resulting balance.

Change PIN: Requires the current PIN, checks that the new PIN contains exactly four digits, and asks the user to confirm it.

Exit: Ends the program with a thank-you message.

8. Testing and Results

I tested the program by trying normal operations as well as incorrect inputs. The checks below describe the expected behaviour of the completed program.

Test	Expected result
Correct PIN	Login succeeds and the menu appears.
Incorrect PIN three times	Access is blocked and the program does not show the ATM menu.
Deposit a positive amount	The amount is added to the balance and recorded in history.
Enter letters instead of an amount	An error message appears and the balance remains unchanged.
Withdraw more than the balance	The program reports insufficient balance and does not deduct money.
Withdraw an amount not divisible by 100	The program asks for an amount in multiples of 100.
View statement before transactions	The program reports that no transactions are found.
Change PIN with mismatched confirmation	The old PIN remains unchanged.
Choose Exit	The program displays a closing message and stops.

9. Limitations and Future Improvements

At the moment, the program keeps its information only while it is running. It has one sample account and a fixed starting balance. The PIN is also stored in the source code, so this version should only be used for learning.

If I continue working on the project, I would like to add file or database storage so that account information is saved between runs. I could also add support for multiple users, dates and times in the mini statement, and stronger security features.

10. Conclusion

Building this ATM Simulator helped me practise the Python basics in a practical way. I learned how to use functions, loops, conditions, lists, dictionaries, and separate modules to build a program with several related features. I also learned that checking user input is important because users may enter unexpected values. Overall, this project gave me useful experience in planning, writing, testing, and organizing a small Python application.

Repository link: To be added after uploading the project to GitHub.

Student note: Please fill in your name and registration number on the cover page, and review the report so it accurately reflects your own work and testing.